UGREEN AI NAS HACKATHON 2026 / ROUND 1

VaultMind AI

Find your files. Review changes before they happen.

01 / What does it do?

VaultMind AI provides evidence-backed answers, explainable smart organization, contextual version comparisons, and approval-first SafeAction planning with cryptographic recovery evidence for personal NAS collections.

02 / Who is it for?

It is built for students, families, researchers and professionals managing years of documents on their NAS. Critical receipts, warranties, policy renewals and project revisions get scattered across folders, making retrieval frustrating and unassisted bulk reorganization risky.

03 / What is my solution?

Users can ask practical questions like, "Find my latest insurance document and explain how it differs from the previous version," or request outcomes such as, "Organize my research papers." VaultMind extracts cited clause deltas, proposes explainable folder structures with confidence ratings, and requires approval before moving files.

04 / How would someone use it?

The user chooses authorized folders and exclusions, then asks a question or organization goal. VaultMind presents citations, diffs, and a preview table showing destination rationales and confidence. Users approve specific numbered actions, and receive a tripartite ledger showing authorized, changed and unchanged items.

05 / What makes it different?

Unlike black-box file organizers, VaultMind emphasizes explainability and deterministic safety. It gives explicit classification rationales, quarantines uncertain documents for manual review, produces cryptographic execution receipts, and enforces ten adversarial safety invariants including stale-plan and collision rejection.

06 / Which NAS features does it use?

The file-agent performs bounded discovery, SHA-256 verification and approved serial moves. The document-agent handles semantic lookup, extraction and version reconciliation. The knowledge-agent exports verified execution receipts and audit summaries when requested.

07 / How is it built?

Authored in native Uliya Markdown orchestration using ug-filesystem, ug-filemgr, file-workflow, ug-document and ug-knowledge through registered platform agents. It embeds no custom Python services or databases in the package, enforces strict exclusion precedence, and halts private processing unless local model execution is confirmed.

08 / What is ready so far?

The Skill passes official static checks with zero errors or warnings. An independent local workbench with 140 synthetic fixtures and 31 passing tests evaluates retrieval, explainable topic classification, version reconciliation, and achieves 100% rejection across ten adversarial safety attack vectors.

09 / How will I test it?

On the assigned NAS, I will validate warranty lookup, multi-year insurance policy diff extraction, explainable research organization, and single-item approved moves. I will also test stale-hash rejection, mid-batch partial failure handling, and separate inverse plan approval.

10 / What are the limits?

Performance relies on UGOS platform agents and local inference latency. Privacy regexes cannot catch all PII, and rollback requires unchanged destination files. Local benchmarks do not prove hardware performance. I used OpenAI Codex to assist with research, orchestration authoring, test suites and documentation.

Third-party disclosure: UGREEN creation kit 0.7.3 and platform experts, under UGREEN terms; used for authoring/validation and runtime respectively. No third-party executable code is included in the Skill ZIP. Local tooling only: Python (PSF), SQLite (public domain), Node.js (MIT), ReportLab (BSD), pypdf (BSD). Sources: ai.ugreen.com/pages/hackathon-starter-kit and ugreenhackathon.devpost.com.